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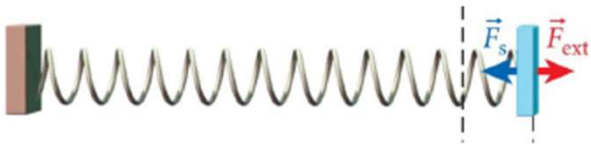
Recap on Work

RECAP: Kinetic energy, work & power

Work is the energy transferred to or from an object due to the action of a force.

- Positive work is a transfer of energy to the object.
- Negative work is a transfer of energy from the object.

Kinetic Energy $K = \frac{1}{2}mv^2$ $\Delta K = K - K_0 = W$ $W = \vec{F} \cdot \Delta \vec{r}$



$$F_s = -kx$$

$$W = -W_s = \frac{1}{2}kx_c^2$$

Work

Power $P = \frac{dW}{dt}$ $\bar{P} = \frac{W}{\Delta t}$ $P = \vec{F} \cdot \vec{v}$

Chapter 8

Conservation of Energy

Law of conservation of energy

*Energy isn't created or destroyed;
it's just shuffled around from one form to another.*

Different forms of energy:

- Kinetic
- Elastic
- Chemical
- Electric
- Thermal
- Nuclear

Consistent with the idea of a “closed system”

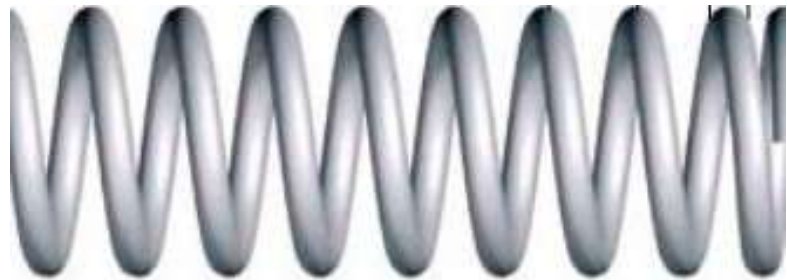
Thought experiment: Dropping things in acid

What would a calorimeter say if we dissolved each spring in a vat of acid?

Would they release the same energy?

Or different?!?!

Uncompressed:



Compressed:



Gravitational potential energy: Hydro



Gravitational potential energy

If there's no energy being created, then where do falling objects get their energy from?

Answer: potential energy

$$U = mgh$$

Where:

m = mass of object

g = gravitational force

h = height above ground

Gravitational potential energy

Conservation of energy means that the amount of energy can't increase or decrease, i.e.:

$$\Delta E = \Delta K + \Delta U = 0$$

When then implies:

$$\Delta K = -\Delta U$$

Which in turn explains why potential and kinetic energy have opposite signs!

Ballistics:

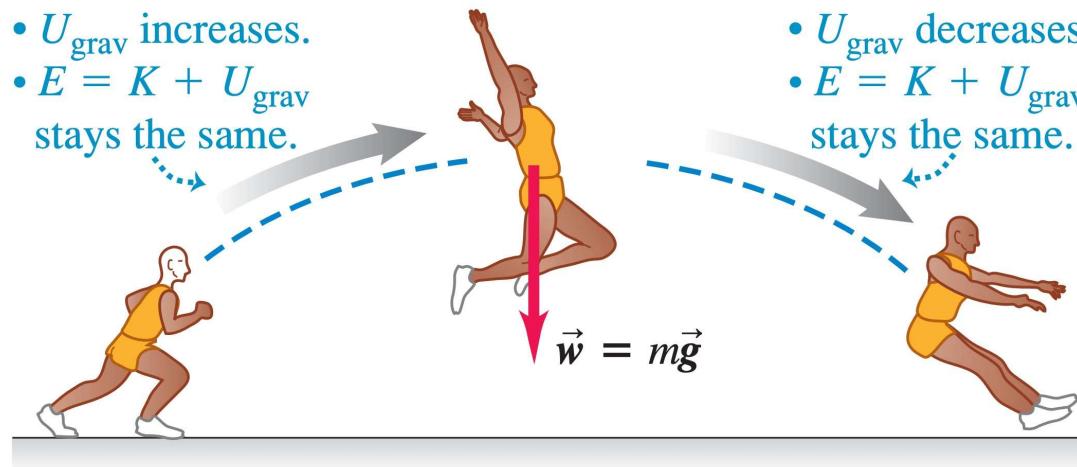
Typically exemplifying involving throwing balls, objects, etc (common in exams!)

Moving up:

- K decreases.
- U_{grav} increases.
- $E = K + U_{\text{grav}}$ stays the same.

Moving down:

- K increases.
- U_{grav} decreases.
- $E = K + U_{\text{grav}}$ stays the same.

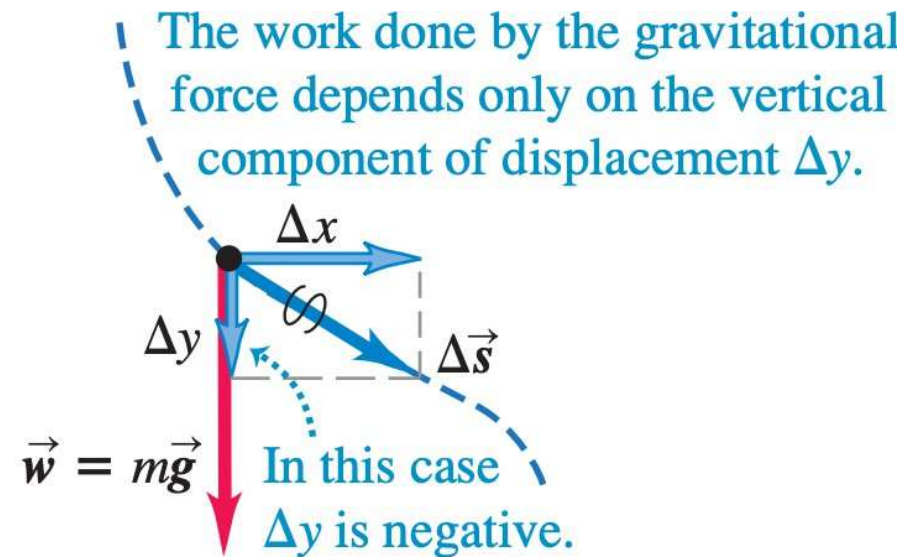
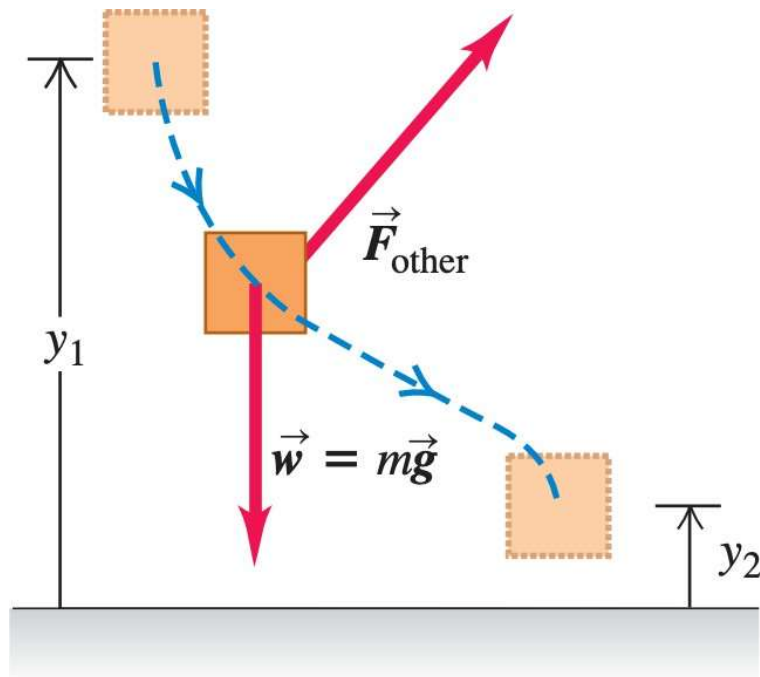


- How does the horizontal velocity change throughout the jump?
- How does the vertical velocity change throughout the jump?
- How does the speed change throughout the jump?

What about curved paths:

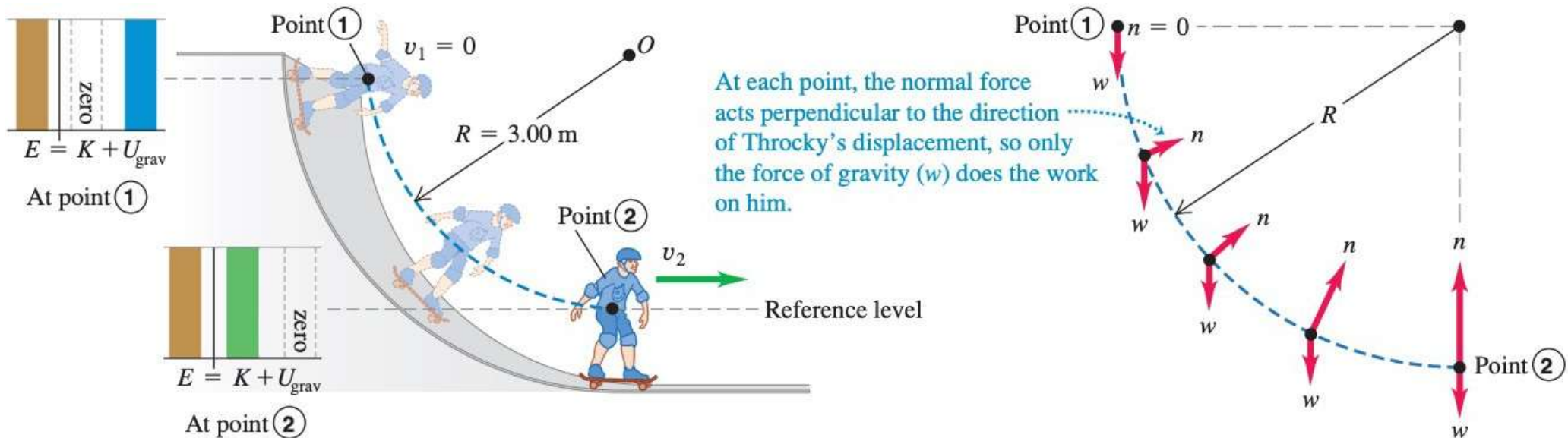


What about curved paths:



Short answer: Consider only the vertical change

Example: Skateboarder



Frictional losses

Friction is the force resisting the motion of an object sliding on a solid surface. The work made the frictional force is:

$$W_f = F_f \Delta s$$

Where:

F_f is the frictional force

Δs is the distance over which that force acts

Friction is the object doing work on its environment, i.e. a negative work !!!

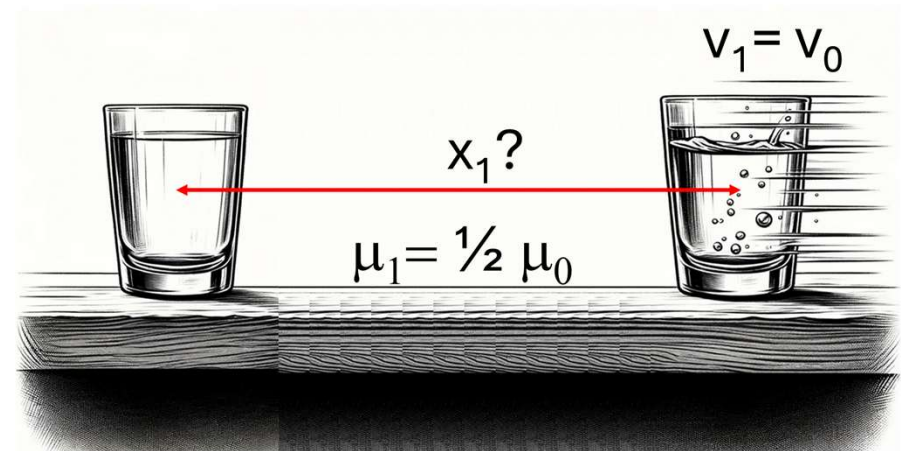
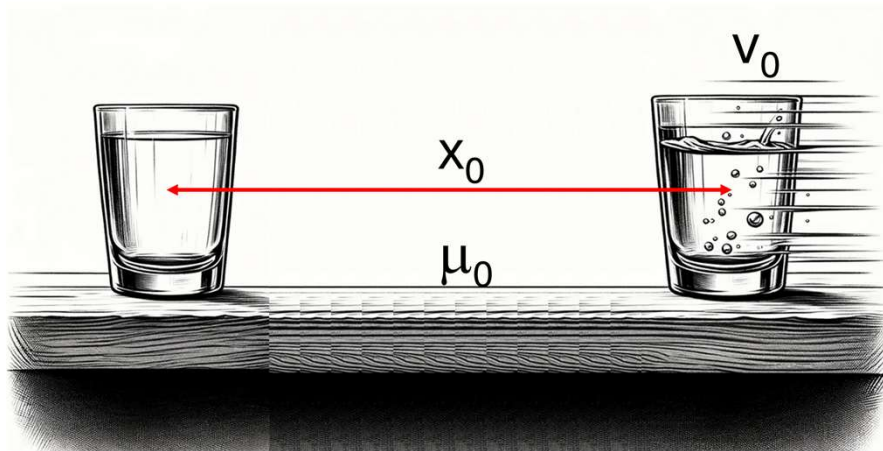
Frictional force is proportional to the normal force from the surface acting on the object: $F_f = \mu_K N$ μ_K is the the coefficient of kinematic friction

Example: Exam Problem May 2023

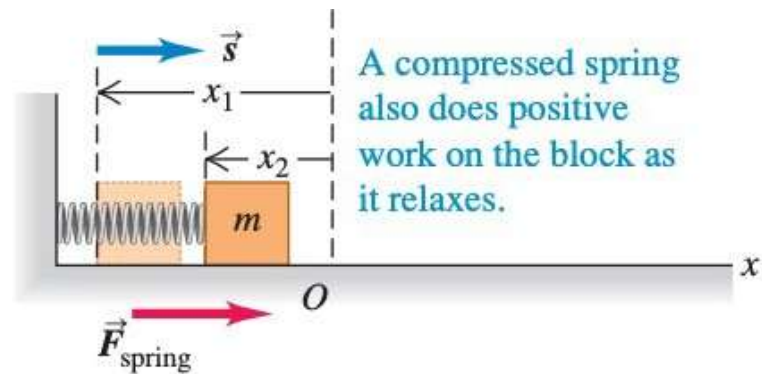
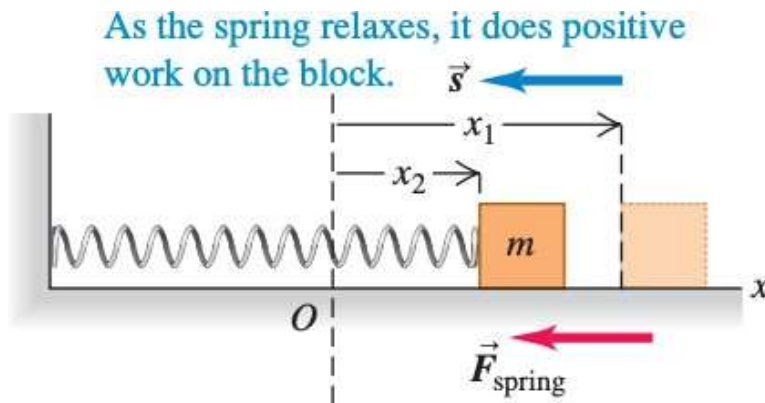
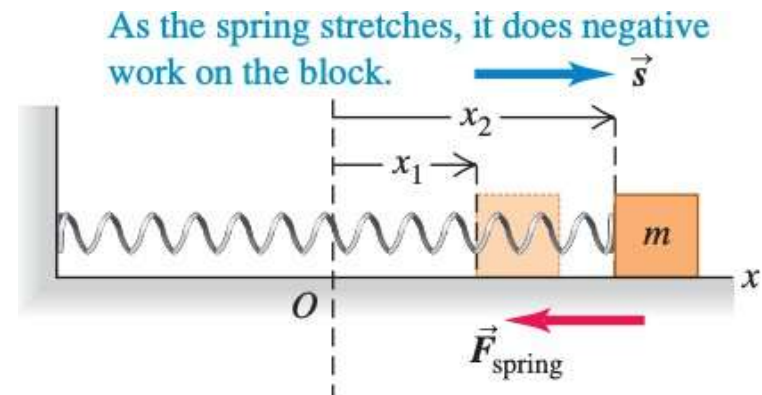
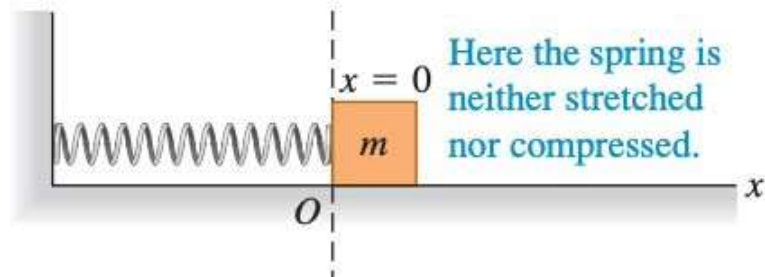
A bartender sends a glass off so that it slides along the counter with an initial speed v . There is friction between the glass and the counter, and the glass stops after a short distance, x . The bartender sends a new glass off at the same speed as before. The glass is similar to the first, but in the meantime, some liquid has been spilled on the bar counter, so the coefficient of kinematic friction is now only half of what it was before.

How far will the glass reach this second time?

Example: Exam Problem May 2023



Elastic potential energy:



Elastic potential energy (formalized)

Elastic potential energy:

$$U_{el} = \frac{1}{2} k x^2$$

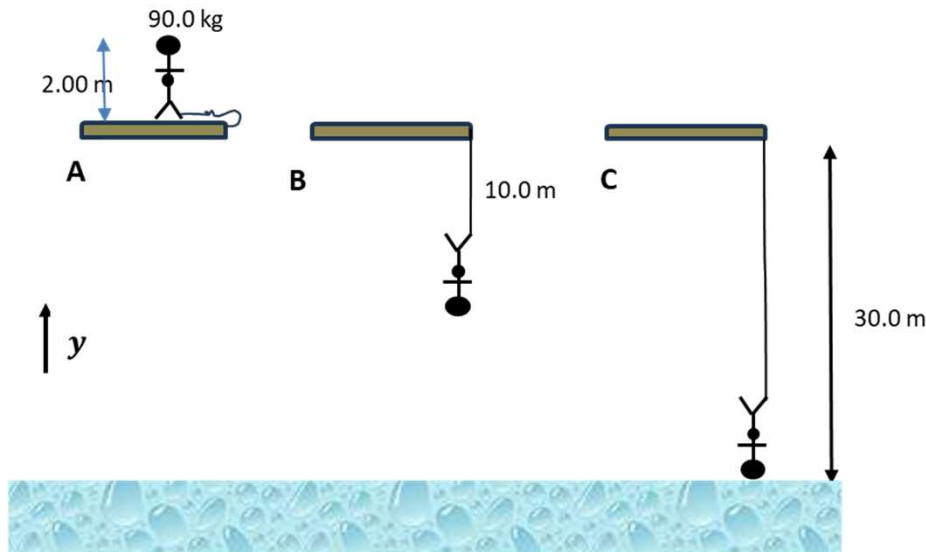
Work done on a spring:

$$W = \frac{1}{2} k (x_2^2 - x_1^2)$$

Work done by a spring:

$$W = \frac{1}{2} k (x_1^2 - x_2^2)$$

Example: Exam Problem August 2023

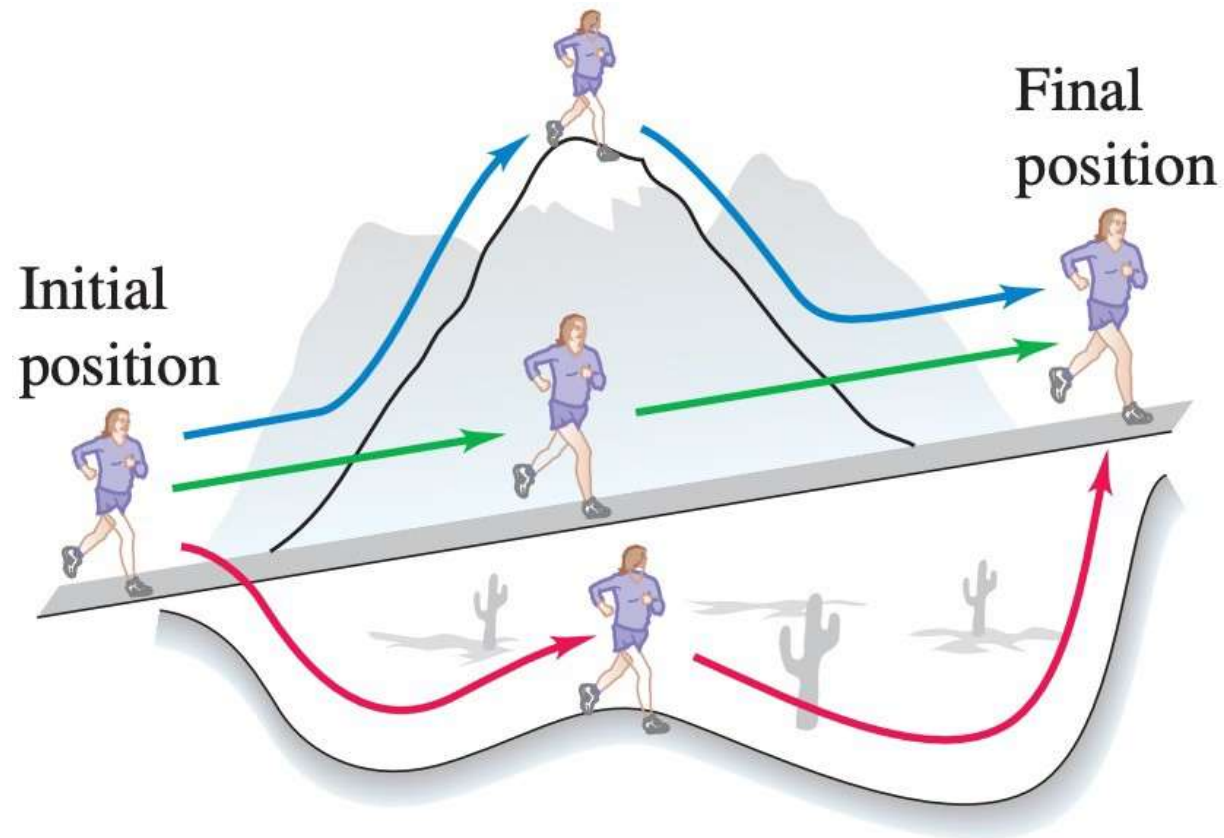


A student with a height of 2.00 meters and a mass of 90.0 kg is on vacation in the south and uses the time to bungee jump from a bridge located 30.0 meters above a river (figure A). The elastic is 10.0 meters long in the unstretched state and has a spring constant, k . The student's center of mass is in the middle of the body and the gravitational acceleration can be taken as $g = 9.82 \text{ m/s}^2$.

What is the student's speed in Figure B just before the elastic begins to stretch?

In figure C, the student's head has just touched the water and luckily her speed here is zero. What is the spring constant?

Conservative vs nonconservative forces



Conservative vs. nonconservative forces

A conservative force is one in which the work is reversible (i.e. there are no losses), and is a difference between initial and final states of a potential energy function (i.e. it's independent of the path taken)

A nonconservative force is essentially everything that's not a conservative force

Example: Gravity is a conservative force, but friction is nonconservative

Force and potential energy:

Because force and potential energy are related: $\Delta U = -\Delta W = -F\Delta x$

We can express force in terms of the potential energy: $F = -\frac{\Delta U}{\Delta x}$

If we then start considering small changes, we can write:

$$F = -\frac{dU}{dx} \qquad U = -\int_{x_1}^{x_2} F dx$$

This means we can describe the energy functions associated with forces.

Energy diagrams

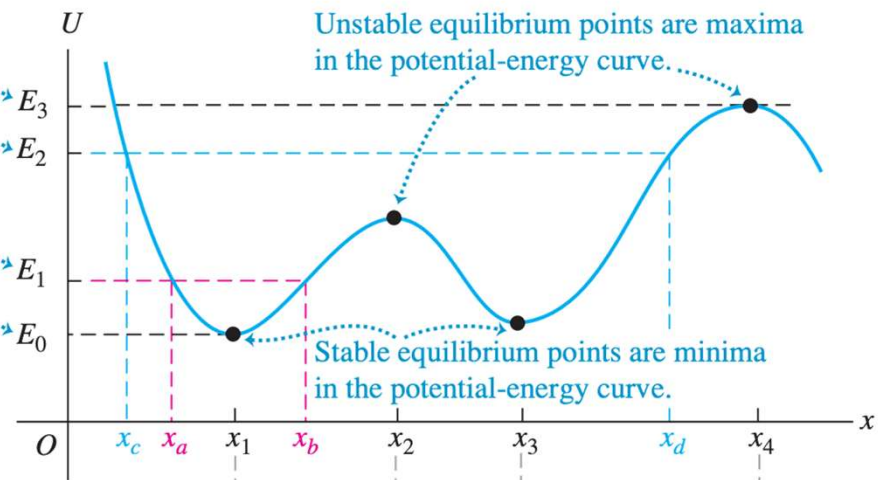
If the total energy $E > E_3$, the particle can “escape” to $x > x_4$.

If $E = E_2$, the particle is trapped between x_c and x_d .

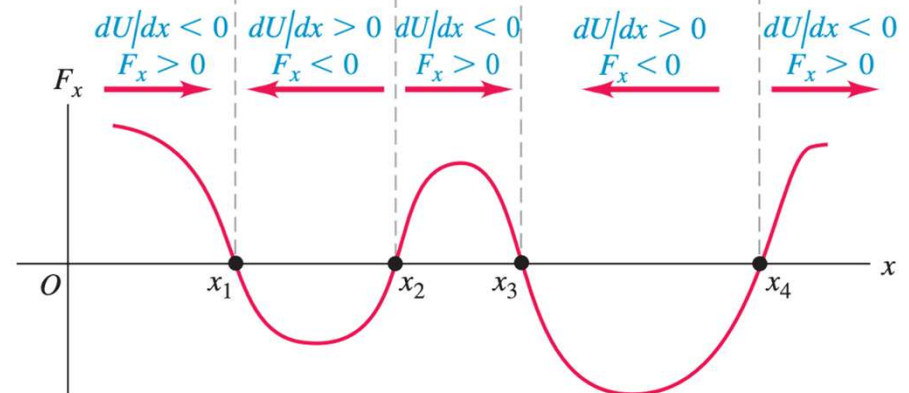
If $E = E_1$, the particle is trapped between x_a and x_b .

Minimum possible energy is E_0 ; the particle is at rest at x_1 .

(a) A hypothetical potential-energy function $U(x)$



(b) The corresponding x -component of force $F_x(x) = -dU(x)/dx$



$$F = -\frac{dU}{dx}$$

Task for next week

Complete the quiz on Work and Energy

Deadline: Sunday 12th